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TrackWise

Feature & Demo Manual

Integrated AICTE Approval Workflow Automation,
Compliance Monitoring and Alert Management System

Smart India Hackathon - Round 2 - Smart Automation

Version 1.1 | 26 October 2026

A team guide: what the product does, how it works, how to run it and how to present it.

All institutions, people, thresholds and figures in the product and in this manual are invented for the demo. They are not real AICTE data or norms.

Repository: github.com/AlexDsouza107/trackwise (public)

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How to use this manual

New to the project? Read sections 1-3, then skim 4. Presenting? Learn section 7 and rehearse it twice. Working on code? Sections 5, 6 and 9 tell you where things live and how to run the tests. Everything here was checked against the code in the repository (v1.1), not written from memory.

1. The 60-second pitch

The problem

Approving an institution, a new course or an intake change is a long chain of hand-offs: documents are checked by hand, compliance is looked at separately from the application, institutions often cannot see where their file is, and nobody is warned when a deadline slips or an approval is about to lapse. Problems are found late, and the same file bounces back for missing paperwork.

What TrackWise does about it

Pillar	What it gives you	Where to see it
Workflow automation	One state machine drives every application. Illegal moves are impossible, every move is audit-logged, and the next owner is notified automatically.	Application detail page, audit trail
Compliance monitoring	Four rules (faculty ratio, infrastructure, accreditation, fees) are scored per institution per month. Compliance is also a hard gate inside the workflow.	Compliance page, heatmap
Smart alerts	Background jobs find SLA breaches, expiring approvals and stale drafts, and escalate them up a ladder with e-mail at higher levels.	Alerts page, escalation rules
Explainable AI	A document check and a risk score. Every point of the risk score is itemised so officers can see why.	AI assessment card

Say it in one breath

> *"TrackWise turns AICTE approvals from a paper chase into a tracked, automatically escalated pipeline: institutions see exactly where they stand, officers see what is most urgent and why it is risky, and administrators are warned before a deadline is missed - with a full audit trail behind every decision."*

Three things judges should remember

- **The rules are in the engine, not in the UI.** A critical compliance failure literally blocks "Compliance cleared"; the screen only shows what the server allows.
- **Nothing happens silently.** Every transition, upload and removal is on the audit trail; every breach raises an alert exactly once, and escalates.
- **The AI explains itself.** No black box - a score of 38 comes with the four reasons that add up to 38.

> **Be upfront about what this is**
>
> This is a working demo MVP with illustrative data. Norms, SLAs and institutions are invented. The AI is rules-based on purpose (explainable) and reads document metadata, not contents. E-mail is a stub. Section 9 lists every limit; stating them yourself earns trust.

2. Who uses it

There are four roles. The server enforces what each may do; the UI simply renders the actions the server says are available. All demo accounts use the password **demo1234**, and the login page offers one-click sign-in tiles.

Role	Demo accounts	What they do
Institution	inst01 ... inst14 @trackwise.demo (inst13, inst14 are *proposed* new institutions)	Create applications, upload and remove documents, submit, respond to deficiencies, track status, see their own alerts and compliance.
Scrutiny officer	officer1, officer2 @trackwise.demo	Work the queue for Document Scrutiny and Compliance Check. Verify documents, review AI and compliance results, clear, raise a deficiency or reject.
Committee member	committee1, committee2 @trackwise.demo	Review applications at the expert-committee stage. Recommend approval or rejection (note required) or raise a deficiency.
Admin	admin@trackwise.demo	Takes the final decision (approve or reject), sees everything, edits escalation rules, runs alert scans, reads the e-mail outbox and analytics.

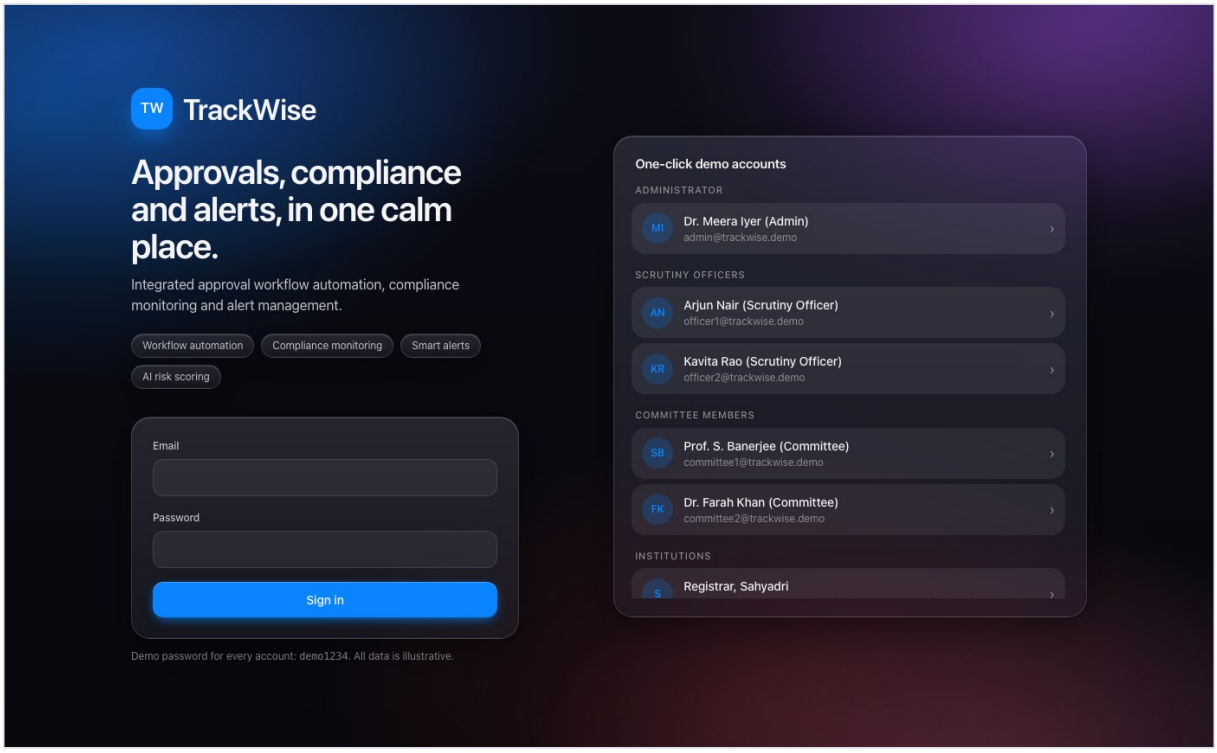
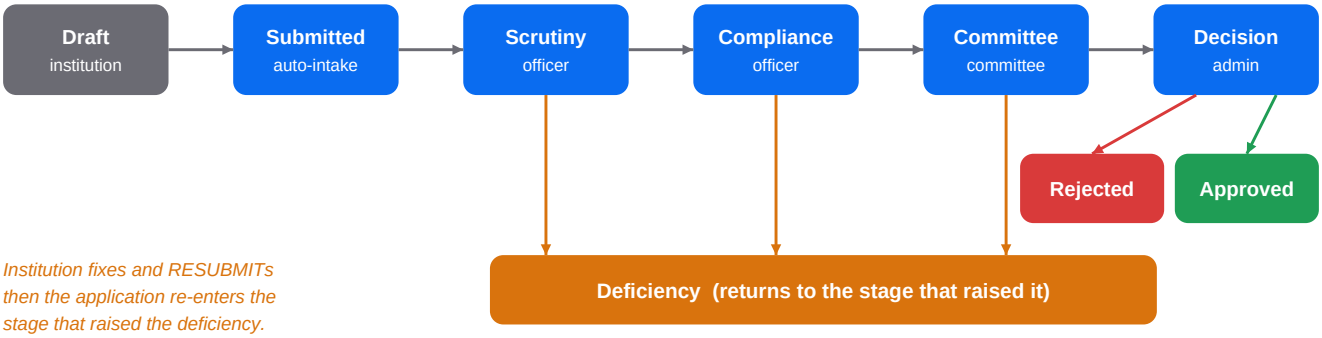


Figure 1 - Login page with one-click demo accounts (dark theme). Each tile signs in instantly.

Tip for presenting: open each role in its own browser profile or private window so you can switch roles without signing out and in.

3. The application lifecycle

Every application is a state machine (XState). The machine says which moves are legal in each state. A separate permission table says which role may trigger each move. Together they make the workflow safe by construction.



Institution fixes and RESUBMITs then the application re-enters the stage that raised the deficiency.

Stages and their SLAs

Stage	Who owns it	SLA (illustrative)	What happens here
Draft	Institution	-	Fill the form, upload the required documents (one per type).
Submitted	System	24 h	Automatic intake: the AI document check and risk score run, then the file moves to scrutiny.
Document Scrutiny	Scrutiny officer	72 h	Check the documents and the AI findings. Verify, or raise a deficiency.
Compliance Check	Scrutiny officer	72 h	Compliance rules are evaluated. Critical failures block clearance.
Expert Committee Review	Committee	120 h	Committee recommends approval or rejection, or raises a deficiency.
Decision	Admin	48 h	Final approve or reject. Approval sets a validity date.
Deficiency	Institution	240 h	The institution fixes the issues and resubmits. The file returns to the stage that raised it.
Approved / Rejected	-	-	Final. Audit trail is complete.

Who can do what

Action (event)	Roles allowed	Note required?	Notes
Submit application (SUBMIT)	Institution	No	Only from Draft.
Documents verified (PASS_SCRUTINY)	Officer, admin	No	Moves to Compliance Check.
Compliance cleared (CLEAR_COMPLIANCE)	Officer, admin	No	**Guard:** blocked while any critical rule fails.
Recommend approval / rejection (RECOMMEND)	Committee, admin	Yes	Two variants carry the outcome.
Approve (APPROVE)	Admin	No	Sets approval validity.
Reject (REJECT)	Officer, admin	Yes	
Raise deficiency (RAISE_DEFICIENCY)	Officer, committee, admin	Yes	Remembers the stage to return to.
Resubmit with corrections (RESUBMIT)	Institution	Yes	Goes back to the stage that raised it.

The deficiency loop in one sentence

An officer or committee member raises a deficiency with a reason; the institution sees a banner and the missing items, uploads fixes and resubmits with a note; the file goes back to the exact stage that raised it, the round counter goes up, and both the stepper and the audit trail show it.

The compliance gate

If a critical rule fails (for example student:faculty ratio), the machine refuses "Compliance cleared". The officer is offered only "Raise deficiency". This is enforced by a guard in the state machine, not by hiding a button.

4. Feature tour

Sign in and welcome

One-click demo tiles or e-mail and password. After signing in, a short welcome sequence ("Welcome, name" with a role-specific line) plays before the dashboard slides in. With reduced-motion enabled in the operating system, the app skips the animation.

Institution portal

- **New application:** choose New Institution, New Course, Intake Increase or Extension of Approval; the form adapts to the type.
- **Documents:** each type has a checklist (for example a new course needs application form, faculty list, lab inventory, curriculum, fee structure). One file per type. **View** opens it in the viewer, **+ Add** uploads, the **x** removes it (only in Draft or Deficiency, only by the owner, and it is audit-logged).
- **Track status:** a stepper shows progress, the SLA chip shows time left or overdue, and the audit trail lists every event with actor, role and note.
- **Deficiency banner:** shows who raised it and why, and lists the missing documents.

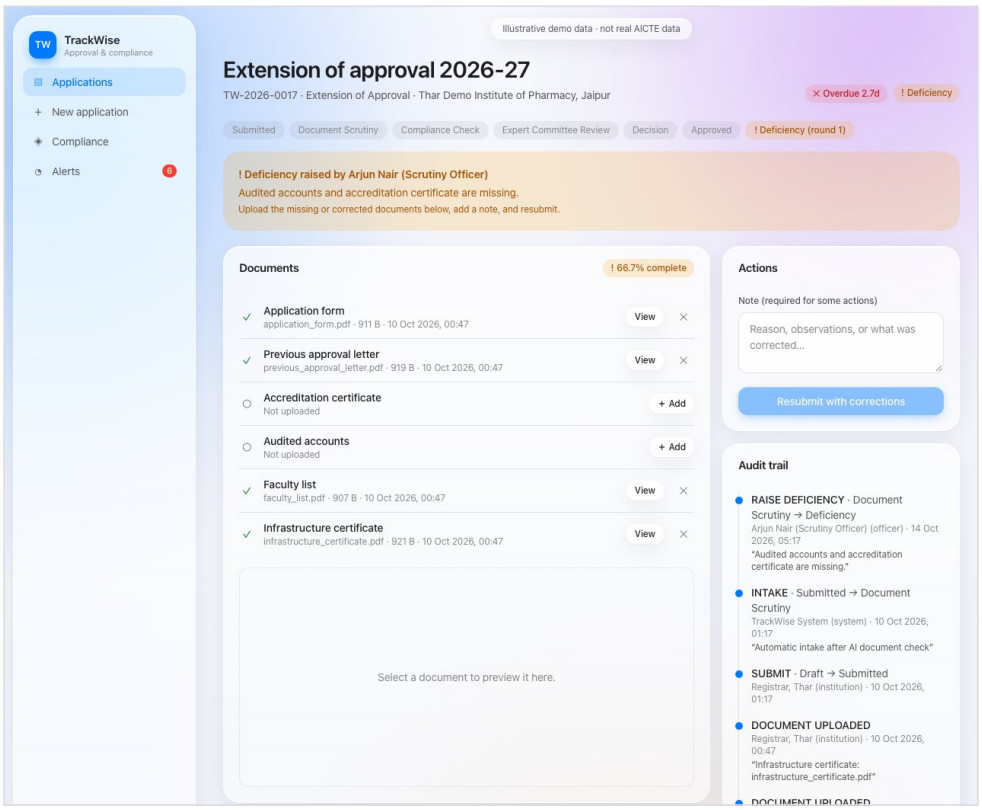


Figure 2 - Institution view of an application in Deficiency: reason banner, checklist with Add/View/remove, resubmit action and audit trail.

Officer and committee queue

The queue shows only the stages a role owns, ordered by SLA urgency, with breached items first. **Queue / All** switches between your stages and everything. Each row shows reference, title, institution, stage, SLA chip, AI risk chip and the number of deficiency rounds.

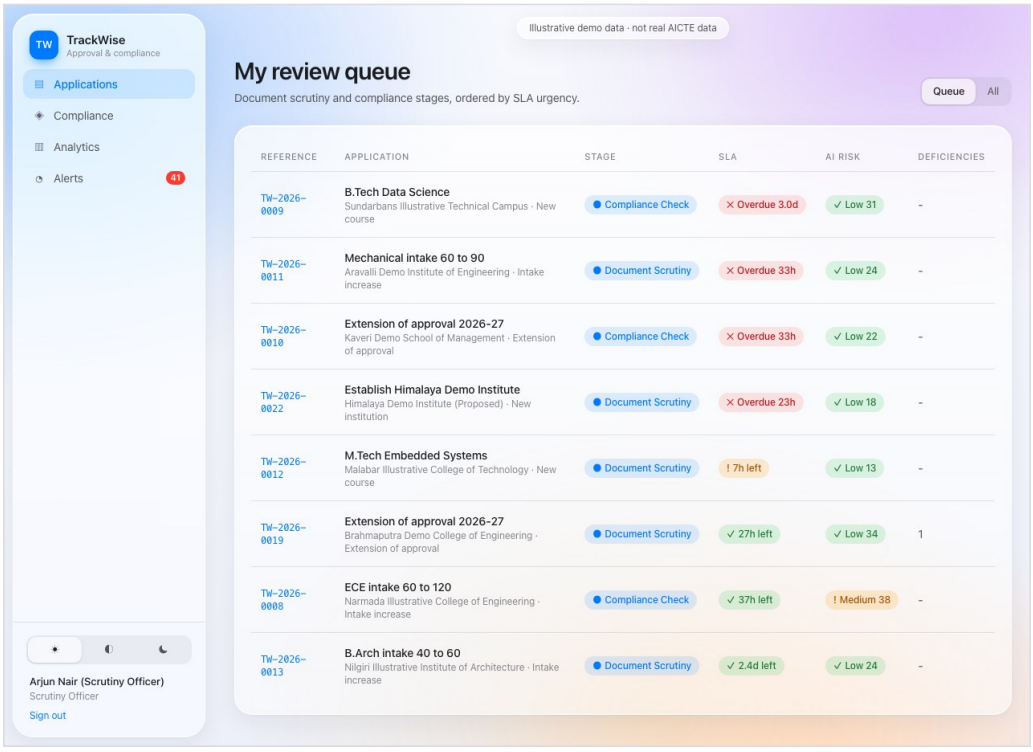


Figure 3 - Scrutiny officer queue. Overdue items pulse and sort first; the AI risk chip pairs a colour with an icon and label.

Application detail

This is the working screen for officers. Left: documents with an inline viewer, application details, the AI assessment and the compliance evaluation. Right: the action panel (only the actions you may take right now) and the audit trail.

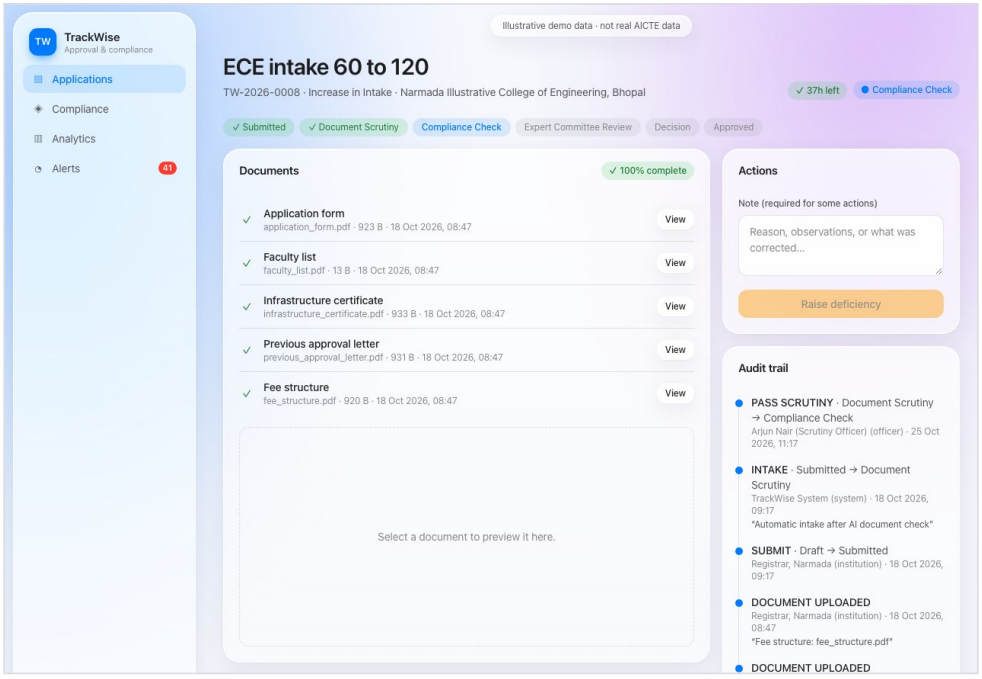


Figure 4 - Application detail: stepper, document checklist and viewer, action panel and audit trail.

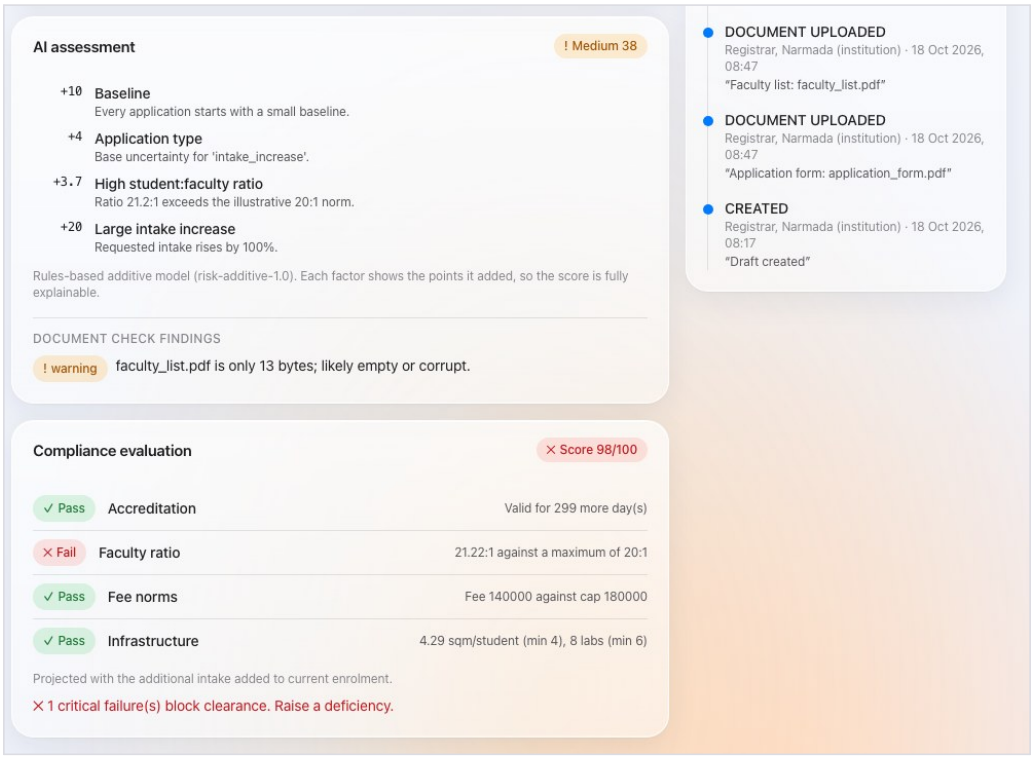


Figure 5 - AI assessment (itemised risk points and document findings) and the compliance evaluation with the blocking message.

Reading the AI assessment

Baseline 10 + application type 4 + high faculty ratio 3.7 + large intake increase 20 = **Medium 38**. Each line says why. The document check separately flagged a 13-byte faculty list as probably empty. The compliance card then shows the critical ratio failure, so the only available action is Raise deficiency.

Compliance monitoring

A heatmap shows every institution against every rule, with a percentage score per cell, the overall score and the change since last month. Click an institution for its six-month trend (with the 70 target line), the latest rule results and the raw numbers (faculty, students, built-up area, labs). **Re-evaluate now** recomputes it.

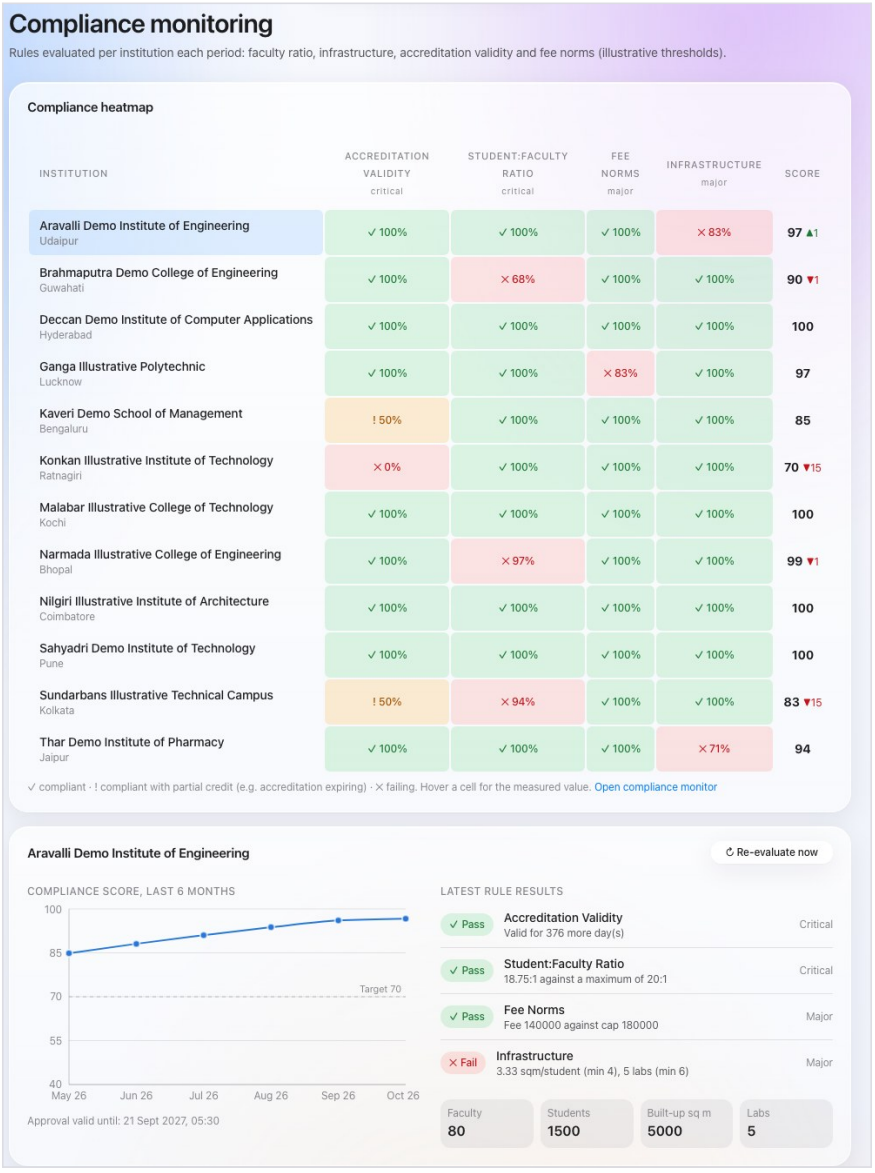


Figure 6 - Compliance heatmap and an institution drill-down. Every cell pairs colour with a symbol and a number.

Alerts and notifications

Four tabs: **My notifications** (inbox with unread count), **Alert log** (every alert with its escalation level), **Escalation rules** (admin edits hours and switches e-mail on or off) and **Email outbox** (the stubbed e-mails). **Run alert scan now** triggers the scan on demand for demos; in the full stack a repeatable BullMQ job runs it on a schedule.

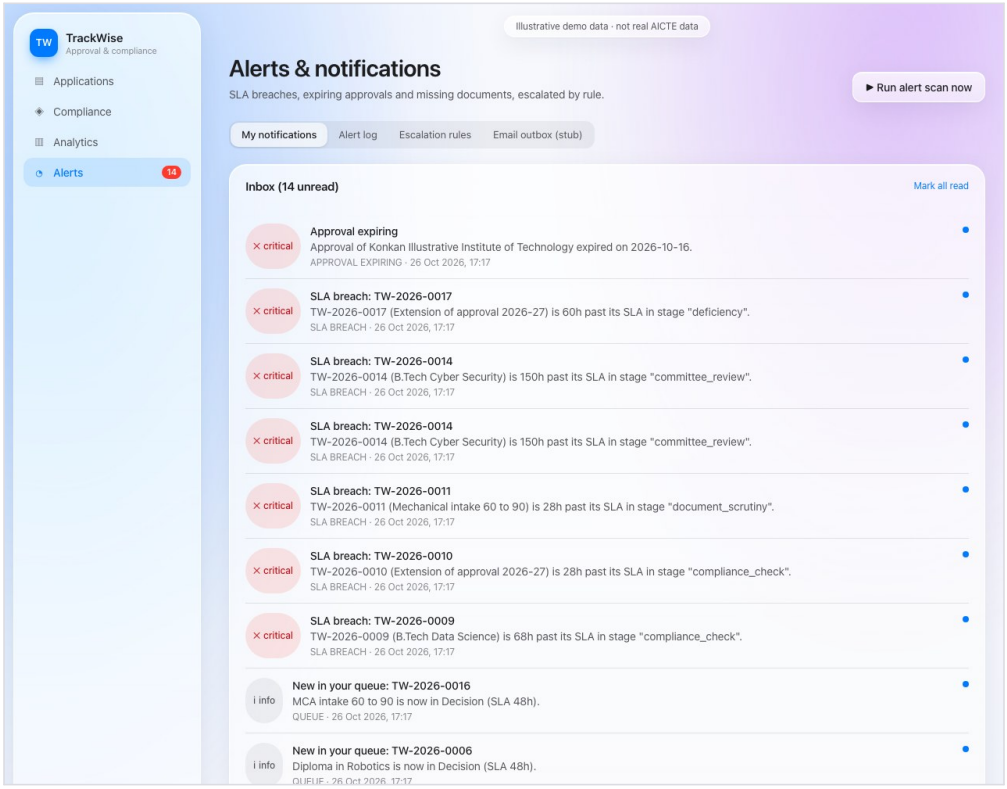


Figure 7 - Admin inbox: escalated SLA breaches and an expired approval, plus queue notifications.

Analytics

Headline tiles (open applications, SLA breached, average turnaround, approved, rejected, alerts raised), pending applications by stage, turnaround by application type, open applications by AI risk level, 14-day alert volume by kind, the average compliance trend and the heatmap.

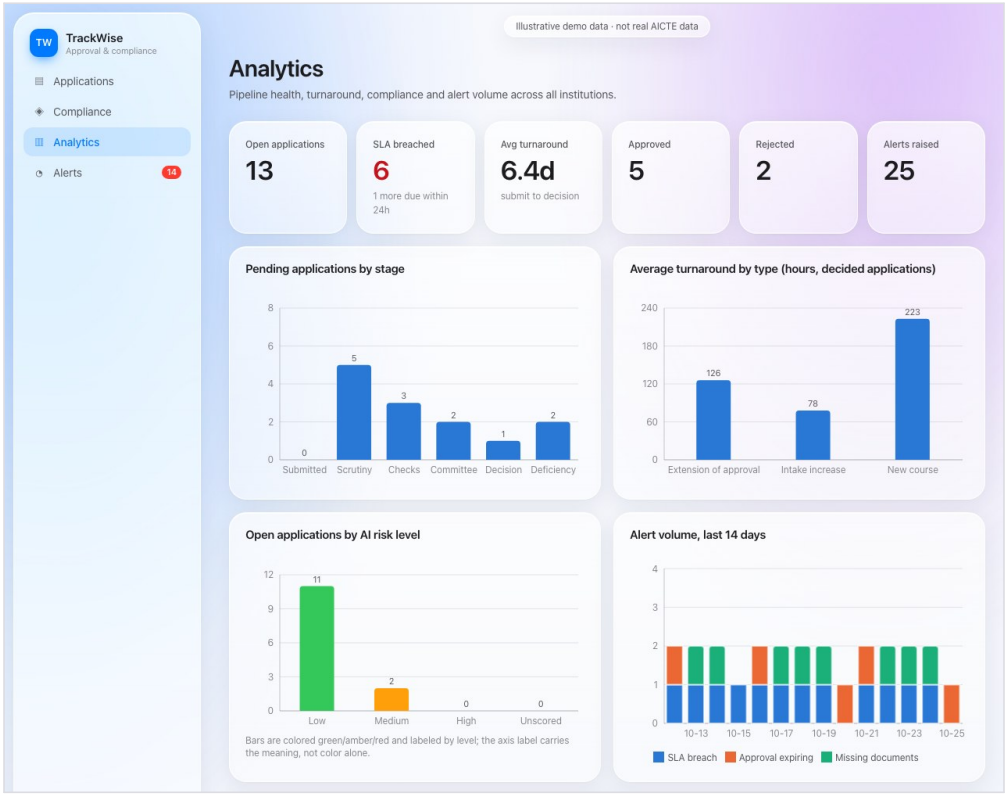


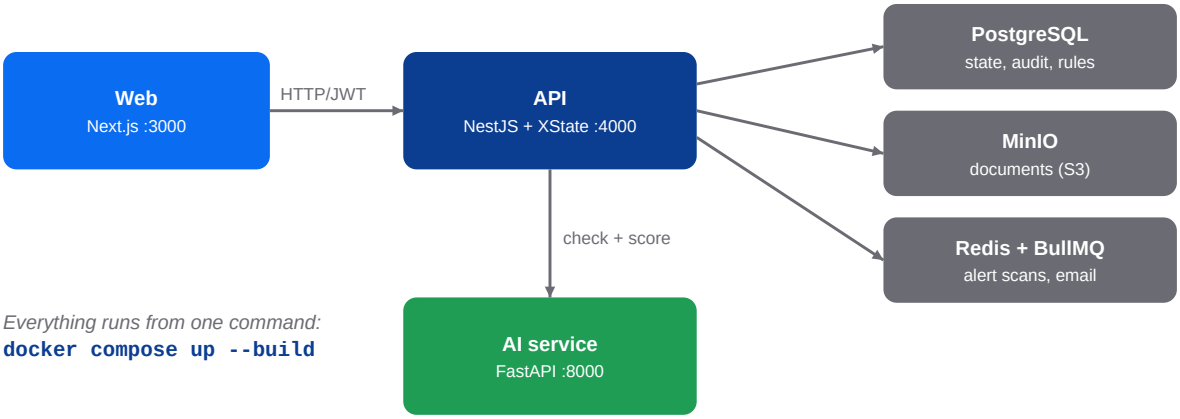
Figure 8 - Analytics dashboard (top half).

Look and feel

- **Liquid-glass design** following Apple conventions: translucent panels, hairline borders, a soft static gradient backdrop and gently drifting colour orbs.
- **Light, Auto and Dark** themes from the toggle at the bottom of the sidebar. Switching plays a day/night dissolve (View Transitions API with a CSS fallback).
- **Motion with purpose:** staggered entrances, count-up numbers, shimmer skeletons while loading, pulsing "Overdue" chips, the sign-in welcome sequence.
- **Accessibility:** status colours are always paired with an icon and a label; all motion is disabled under "reduce motion"; a reduced-transparency fallback replaces the blur.
- **Demo banner:** "Illustrative demo data - not real AICTE data" is always visible at the top of every screen.

5. How it works under the hood

Architecture



Everything runs from one command:
`docker compose up --build`

Layer	Technology	Responsibility
web/	Next.js 14, React 18, Tailwind, Recharts	All screens. Renders the actions the server returns; charts use a colour-blind-checked palette.
api/	NestJS 11, XState 5, zod, JWT	Auth and RBAC, workflow engine, compliance engine, alerts, analytics, seeding.
ai/	FastAPI, pydantic v2	/check-documents and /risk-score. No ML dependencies; rules-based and explainable.
Data	PostgreSQL 16	Applications, documents, audit log, compliance results, alerts, notifications, escalation rules, e-mail outbox.
Files	MinIO (S3 API)	Uploaded documents; streamed back through the API so access control stays in one place.
Jobs	Redis + BullMQ	Repeatable alert-scan job and e-mail jobs, with workers running in the API process.

AI service

Document check looks at metadata only (type, size, extension, hash) and returns a completeness percentage, the missing types and findings:

Finding	Severity	Meaning
MISSING_DOCUMENT	critical	A required type has no file.
UNSUPPORTED_FORMAT	warning	Extension is not an allowed document or image type.
TINY_FILE	warning	File is implausibly small (likely empty or corrupt).
DUPLICATE_CONTENT	warning	The same file was uploaded for several document types.
UNEXPECTED_DOCUMENT	info	A document type that this application type does not need.

Risk score is additive: a baseline of 10 plus itemised factors, capped at 100. Level: below 35 is Low, below 65 is Medium, otherwise High.

Factor	Points
Application type base	New institution 8, intake increase 4, new course 3, extension 0
Missing documents	8 each, capped at 32
Low compliance score	25 if below 50, 15 if below 70, 6 if below 85

Factor	Points
High student:faculty ratio	3 per point above 20:1, capped at 20
Large intake increase	10 if above 25%, 20 if above 50%, 28 if above 100%
Fee near or above cap	4 if above 90% of cap; more if over the cap, capped at 20
Accreditation validity	20 if lapsed, 8 if under 180 days left
Prior deficiencies	5 each, capped at 15

If the AI service is down, the application still moves and the panel says the assessment was unavailable. The AI never blocks the workflow.

Compliance engine

Each rule returns pass or fail, a score between 0 and 1 (partial credit), the measured value and the threshold. The institution score is the weight-averaged rule score times 100. Critical rules carry weight 3 and major rules weight 2. The trend chart marks a target of 70.

Rule	Severity	Weight	Illustrative threshold
Student:faculty ratio	critical	3	No more than 20 students per faculty member
Infrastructure	major	2	At least 4 sq m built-up area per student and at least 6 labs
Accreditation validity	critical	3	Must be valid; partial credit when under 180 days remain; 0 when lapsed
Fee norms	major	2	Annual fee not above the cap (illustrative)

For intake-increase applications, compliance is evaluated as a what-if: the extra seats are added to current enrolment, so a request that would break the faculty ratio is caught before approval.

Alerts and escalation

A scan finds three kinds of problem. Each (entity, level) pair raises **once** (deduplicated by key), so running the scan twice never spams anyone. Hours mean different things per kind: SLA breach = hours past due; approval expiring = hours *before* expiry; missing documents = hours the application has waited in its stage.

Kind	Level	After (hours)	Notifies	E-mail?
SLA breach	1	0	Owner of the current stage	No
SLA breach	2	24	Admin	Yes
SLA breach	3	72	Admin	Yes
Approval expiring	1	2160 before (90 days)	Institution	No
Approval expiring	2	720 before (30 days)	Institution	Yes
Approval expiring	3	720 before (30 days)	Admin	No
Missing documents	1	24	Institution	No
Missing documents	2	96	Institution	Yes
Missing documents	3	168	Officer	No

E-mail is a stub: a row in the outbox table and a BullMQ job that logs "[EMAIL STUB]". Rules are editable by the admin in the UI.

Design decisions worth knowing

- **The machine decides validity; the API decides permission.** The UI renders `availableActions` from the server, so buttons cannot drift from the rules.
- **Transitions are transactional.** The row is locked (`SELECT ... FOR UPDATE`), the audit row is written, then entry side effects run (AI refresh, compliance what-if, notifications, approval validity). They can chain further

events - that is how the automatic intake after Submit works.

- **Seed data goes through the real engine** with back-dated timestamps, so overdue items, escalations and audit trails are genuine rather than hard-coded.
- **One document per type.** Uploading again replaces the old file in the database and in storage; removal is audit-logged.
- **Documents are streamed through the API** (no presigned links), keeping RBAC in one place.

6. Running and testing it

Option A - full stack with Docker (the intended way)

```
docker compose up --build
```

Service	URL	Notes
Web app	http://localhost:3000	Sign in with any demo account
API	http://localhost:4000/health	Health and record count
AI service docs	http://localhost:8000/docs	Swagger UI for /check-documents and /risk-score
MinIO console	http://localhost:9001	Demo credentials are in the README

On first boot the API creates the schema and seeds the demo data in about two seconds. Watch `docker compose logs -f api` for "[EMAIL STUB]" lines.

> **Honest status of the Docker path**
>
> The compose file has **not been run end-to-end yet** because the machine used for development had no Docker installed. Every service was verified natively (Option B). Whoever has Docker should run it once before the presentation and fix anything that appears.

Option B - without Docker (verified)

You need Node 20+, Python 3.11+ and one PostgreSQL database. MinIO and Redis are replaced by in-memory stand-ins, so uploaded documents disappear on restart and alert scans are triggered with the "Run alert scan now" button.

```
DATABASE_URL=postgres://user@127.0.0.1:5432/trackwise ./infra/dev-lite.sh
```

Resetting the demo data

Your code is safe in git. In no-Docker mode the database persists but documents do not; to start fresh, drop and recreate the database and start the API again (it reseeds). In Docker, `docker compose down -v` removes the volumes.

Tests

Suite	Command	Covers
AI service (3)	cd ai && .venv/bin/python -m pytest	Completeness, findings, risk explanation adds up
API unit (7)	cd api && npm test	Compliance engine, workflow machine and guards
API end-to-end (11)	cd api && npm run test:e2e	Seed, happy path, deficiency loop, compliance gate, RBAC, document removal, alert scan, analytics

The end-to-end suite uses real Postgres and the real AI service, with MinIO and Redis replaced by in-memory fakes. Set `DATABASE_URL` and `AI_URL` first (see README).

7. The demo script

Total about 8 minutes. Seeded applications are found by title; reference numbers look like TW-2026-0008 and depend on when the data was seeded. Lines in blue are suggested things to say.

Before you start

- Three browser profiles or private windows: one for institution, one for officer or committee, one for admin.
- Check `/health` returns a number of applications, and that you can sign in.
- Do one dry run so nothing about the first load surprises you.

A. Happy path (about 3 minutes)

Step	Sign in as	Do this	Say this
1	inst01	New application, New Course. Title "B.Tech Quantum Computing", UG, intake 30. Create draft.	"An institution starts a request in under a minute."
2	inst01	Upload all five documents (any small PDF or image); watch the checklist tick. Submit.	"On submit, the AI checks the documents and scores risk automatically, and the file moves into scrutiny."
3	officer1	Open the queue (breached items first). Open your application. View a document, point at the AI assessment. Click Documents verified, then review rules, then Compliance cleared.	"The officer sees the most urgent work first, and exactly why something is risky."
4	committee1	Open it, Recommend approval with a note.	"The committee stage needs a recorded reason."
5	admin	Approve. Open the audit trail.	"Every step is on the record: who, which role, when, and why."
6	inst01 / admin	Alerts shows the status notifications. Admin: Alerts, Email outbox.	"The institution is notified at each step; e-mails are stubbed for the demo."

B. Deficiency loop (about 2 minutes)

Step	Sign in as	Do this	Say this
1	inst05	Open "Extension of approval 2026-27" (Thar Demo Institute of Pharmacy). It is in Deficiency and overdue; read the banner.	"The institution sees exactly what is wrong and what is missing."
2	inst05	Add the Accreditation certificate and Audited accounts, write a note, Resubmit with corrections.	"One click sends it back - to the same stage that raised the problem."
3	officer1	The file is back in the queue with Deficiencies = 1.	"Round counts are tracked and feed the risk score."
4	any institution	Optional live version: create an application, upload only some documents, submit, then as officer click Raise deficiency (it refuses without a note).	"Notes are mandatory where accountability matters."

C. Compliance gate, alerts and analytics (about 3 minutes)

Step	Sign in as	Do this	Say this
1	officer1	Open "ECE intake 60 to 120" (Narmada). Compliance shows a critical ratio failure. Only Raise deficiency is offered.	"Critical failures are enforced by the state machine, not by hiding buttons."
2	officer1	Compliance page: heatmap, then click Konkan (lapsed accreditation) for the trend and rule detail.	"Compliance is monitored continuously, not just at application time."
3	admin	Alerts: Alert log (SLA breaches with levels), Escalation rules (edit hours), Run alert scan now, run it a second time.	"Three levels of escalation. The second scan raises nothing - alerts are deduplicated."
4	admin	Analytics: pending by stage, turnaround, risk mix, alert volume, heatmap.	"This is the picture an administrator needs to see bottlenecks."

If something goes wrong on stage

Refresh once. If a document will not open, say that uploaded files live in memory in the no-Docker mode and use a seeded application instead. Keep a screen recording of the full demo as a backup.

8. Questions judges will ask

Is the AI real?

It is a transparent, rules-based model by design: officers can see every factor behind a score, which matters for a public-sector approval process. It does not read document contents yet. The next step is OCR or LLM extraction and cross-checking fields against the form, then a trained model whose outputs keep the same itemised explanation.

Why a state machine?

It makes illegal transitions impossible and gives a single place to define the process. Changing the approval process means editing one file, and the tests prove the guards, such as the compliance gate.

How is it secure?

JWT authentication, server-side RBAC on every endpoint, ownership checks (an institution cannot open another institution's file - the API returns 403), validated input (zod) and documents served only through the API. Demo limits: shared passwords, tokens in local storage, no rate limiting yet.

Is the data real?

No. Institutions, people, norms and SLAs are invented and marked as illustrative in the app and the README. We chose not to imply real AICTE figures.

Does it scale?

Stateless API and web tiers, PostgreSQL for state, queue-based background work. We have not load-tested it; that is on the roadmap.

How would it plug into real systems?

Through adapters at the edges: identity and document sources (for example DigiLocker-style fetch), accreditation lookups, real e-mail or SMS gateways. The e-mail stub is already behind a queue job so swapping in SMTP is local to one worker.

What happens if the AI service is down?

The workflow continues; the AI panel says the assessment is unavailable. The AI never blocks an application.

How do you prevent duplicate alerts?

Every alert has a dedupe key built from the entity and the escalation level, and it is unique in the database, so rescanning is safe.

What did you test?

3 AI tests, 7 unit tests and 11 end-to-end tests against a real database and the real AI service. The Docker composition itself is the one thing not yet run end-to-end.

9. Limits, roadmap and repo map

What is a stand-in

Area	Today	Production would use
Norms and SLAs	Illustrative placeholders	Official AICTE norms, configurable per approval type
AI	Rules-based, metadata only	OCR / LLM extraction plus a trained model with explanations
E-mail	Outbox table and a logging job	SMTP or an SMS/WhatsApp gateway
Auth	Demo JWT, shared passwords, local storage	SSO, httpOnly cookies, refresh tokens, rate limiting
Documents	Metadata checks, streamed through API	Virus scan, content parsing, retention policy
Intake projections	Extra seats added once to current enrolment	Year-by-year intake model
Docker stack	Written but not yet run end-to-end	Verified in CI on every push

Roadmap ideas, in priority order

- Verify `docker compose up` on a Docker machine and deploy a public demo URL.
- Document intelligence: extract faculty count, land area and dates; flag mismatches with the form.
- Auto-drafted deficiency letters and an institution-side pre-submission checker.
- Real e-mail/SMS, a public status tracker, multilingual UI, tamper-evident audit log, load testing and CI.

Repo map

Where	What to look for
api/src/workflow/machine.ts	States, events, guards, SLA hours, the ACTIONS (role) table
api/src/workflow/applications.service.ts	Transition dispatch (transaction and side effects), documents, audit log
api/src/compliance/engine.ts	The four rules and the score formula
api/src/alerts/scanner.service.ts	Default escalation ladder and the scan
api/src/seed/seed.service.ts	Illustrative institutions and applications, seeded through the real engine
api/test/e2e.spec.ts	End-to-end scenarios
ai/app/risk.py, documents.py	Risk factors and document findings
web/app/(app)/...	Screens: applications, compliance, analytics, alerts
web/app/globals.css	Design tokens, glass utility, motion and theme transition
docker-compose.yml, infra/dev-lite.sh	Full stack, and the no-Docker runner
README.md	Run instructions and the same demo script in short form

Glossary

Term	Meaning
SLA	Service-level agreement: the time a stage is allowed to take. Past it, the application is "overdue" and an alert fires.
Deficiency	A formal request for the institution to fix or add something before the file can continue.
Escalation	Repeating an alert to a more senior role as the delay grows (levels 1 to 3).
Guard	A condition on a state-machine transition; if it is false the transition is refused.
Audit trail	The immutable list of events on an application: who, which role, when, from which state to which, and the note.
Dedupe key	A unique key that stops the same alert being raised twice.